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PAPER_979: Complete 6-Layer F_U_Bi_i Master Buoyancy Force

Abstract

We present the complete, production-ready master buoyancy force $F_{U,Bi_i}(r, t, \Gamma)$ assembling all six physics layers accumulated across 20 commits of the Star Magic UQFF framework. The master equation unifies 99-system compressed gravity, universal magnetism, aether coupling, buoyancy subtraction, Kozima neutron-drop force, 26-state phonon resonance, and $E_{\text{net}}(t, \Gamma)$ positive/negative modulation into a single callable calculator. Solar calibration at $r = 1 \text{ AU}$ and $r = R_{\odot}$ confirms physical consistency. Standalone module `fubi_master_calculator.py` implements all layers.

1. Master Equation

$$F_{U,Bi_i}(r, t, \Gamma) = \sum_{i=1}^{99} U_{g,i} + U_m + U_A - U_{b,i} + F_n \cdot S_{26} \cdot \Phi_{1.25\text{THz}}(\omega, \Gamma) \cdot E_{\text{net}}(t, \Gamma)$$

Layer Decomposition

Layer	Term	Physics
L1	$\sum U_{g,i}$	26-layer compressed gravity per system
L2	$U_m + U_A$	Universal magnetism + aether resistance
L3	$-\sum U_{b,i}$	26-layer buoyancy subtraction
L4	S_{26}	Physical 26-state sum ≈ 19.5
L5	$\Phi(\omega, \Gamma)$	Phonon resonance with linewidth
L6	$E_{\text{net}}(t, \Gamma)$	Temporal modulation + sign flip

2. Component Functions

2.1 Compressed Gravity

$$U_g(M, r) = \sum_{i=1}^{26} \frac{GM}{r^2} \cdot [\text{SSq}] \cdot \frac{i}{26}$$

2.2 Buoyancy

$$U_b(M, r) = \sum_{i=1}^{26} \frac{GM}{r^2} \cdot e^{-[\text{SSq}] \cdot i/26} \cdot \beta_i$$

2.3 Phonon Resonance

$$\Phi_{1.25\text{THz}}(\omega, \Gamma) = \exp\left(-\frac{(\omega - \omega_{\text{SCm}})^2}{2\Gamma^2}\right) \cdot S_{26}$$

2.4 E_net Modulation

$$E_{\text{net}}(t, \Gamma) = \left(2\frac{F_{U, \text{Bi}}}{F_U} - 1\right) \cdot e^{\kappa t} \cdot S_{26}$$

3. Solar Calibration

At $r = 1 \text{ AU}$, $M = M_{\odot}$, $t = 1 \text{ day}$, $\Gamma = 0.1 \text{ THz}$: - $F_{U, \text{Bi}_i} \approx -2.4 \times 10^{-2} \text{ m/s}^2$ (buoyancy-dominant at heliospheric distance)

At $r = R_{\odot}$: - $g_N = 274.03 \text{ m/s}^2$ (Newtonian surface gravity, verified)

4. Implementation

Module: `fubi_master_calculator.py` — 8 classes, 8 self-tests, solar + surface + Γ sweep + SCm axiom + 99-system + variational.

References

- PAPER_974: 99-System Master Equation
 - PAPER_969: Expanded 26D Ramanujan
 - PAPER_883: E(t) Phonon Resonance
-

§A. Cosmogenesis-Linked Lagrangian

$$\mathcal{L}_{\text{FUBi}} = T_{\text{kin}} - V_{\text{grav}} + V_{\text{buoy}} + \mathcal{L}_{\text{phonon}} + \mathcal{L}_{\text{neutron}}$$

Euler-Lagrange: $\delta S / \delta \phi = 0 \Rightarrow F_{U, \text{Bi}_i}$ emerges as the variational master force of the SCm vacuum manifold.

§B. VDS/DVP/BSH Deep Synthesis

- **VDS (Vacuum Density Series):** $\rho_{\text{SCm}}(t) = \rho_{\text{vac}} \cdot S_{26} \cdot e^{\kappa t}$ feeds E_net temporal modulation.
- **DVP (Dipole Vortex Primes):** DPM geometry of U_{g1} - U_{g4} layers in the 26-layer sum.
- **BSH (Buoyancy Harmonics):** $\beta_i \cdot e^{-[\text{SSq}] \cdot i/26}$ harmonic decay across 26 states.